OMEPRAZOLE

Before breakfast

Reduces clopidogrel effect

CHEMICAL NAME (RS)-5-methoxy-2-{[(4-methoxy-3,5-dimethylpyridin-2-yl)methyl]sulfinyl}-1H-benzimidazole

CLASS Proton pump inhibitor (PPI)

TRADE NAMES Losec · Prilosec · Omez · Risek

THERAPEUTIC USES / INDICATIONS

- Peptic ulcer (gastric, duodenal) treatment and prevention (with NSAIDs).
- Gastro-oesophageal reflux disease (GORD), oesophagitis.
- *H. pylori* eradication (triple therapy).
- Zollinger–Ellison syndrome; dyspepsia; upper GI bleeding (IV after endoscopy); stress-ulcer prophylaxis.

ADMINISTRATION

- **PO** 30–60 min before breakfast; swallow capsules whole (or open and sprinkle on yogurt; MUPS dispersible).
- **IV** injection over 5 min or infusion over 20–30 min.
- NG tube: use dispersible tablets or suspension.

DOSAGES

- **GORD / ulcer healing:** 20–40 mg once daily for 4–8 weeks.
- **Maintenance / NSAID prophylaxis:** 10–20 mg once daily.
- **H. pylori:** 20 mg twice daily with 2 antibiotics for 7–14 days.
- **Zollinger–Ellison:** 60 mg daily, up to 120 mg/day.
- **Children:** 0.7–1.4 mg/kg once daily (max 40 mg).

PHARMACOKINETICS

- **Onset:** 1 h; max effect after 3–5 days. **Duration:** ~72 h (irreversible pump binding).
- Acid-labile (enteric-coated); metabolised by CYP2C19 and 3A4.
- Excreted in urine. **t½:** 0.5–1 h.

PHARMACODYNAMICS

- Prodrug activated in acid → irreversibly inhibits H^+/K^+-ATPase (proton pump) in gastric parietal cells → ↓ basal and stimulated acid secretion by up to 90%.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Headache, diarrhoea, abdominal pain.
- Nausea, flatulence.
- Dizziness.

ADVERSE EFFECTS (SERIOUS)

- *C. difficile* infection; pneumonia.
- Hypomagnesaemia (long-term), B12 deficiency.
- Osteoporotic fractures (long-term).
- Acute interstitial nephritis; subacute cutaneous lupus; masking gastric cancer symptoms.

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity to PPIs.
- Use with nelfinavir/atazanavir.
- **Caution:** alarm symptoms (weight loss, dysphagia, GI bleeding, age >55) — endoscopy first.

KEY INTERACTIONS

- **Clopidogrel:** ↓ activation (CYP2C19); use pantoprazole instead.
- **Methotrexate (high dose):** ↑ levels.
- **Ketoconazole, itraconazole, atazanavir:** ↓ absorption.
- **Warfarin, phenytoin, diazepam:** ↑ levels.

NURSING MANAGEMENT

- Give before breakfast.
- Monitor symptoms and signs of GI bleeding (melaena, Hb).
- Long-term: magnesium, B12, review continued need (deprescribe).
- Check for clopidogrel.

PATIENT EDUCATION

- Take 30–60 min before breakfast; swallow whole.
- Avoid trigger foods, alcohol, smoking; raise head of bed for reflux.
- Report black stools, vomiting blood, difficulty swallowing, weight loss.
- Do not use long-term without review.

KEY POINTS

- Irreversible proton pump inhibitor.
- Avoid with clopidogrel (use pantoprazole).
- 20–40 mg before breakfast.
- Long-term: low Mg, B12, fractures, C. diff.

SOURCES CHECKED **[500]** Omeprazole 1 mg/ml Powder for Oral Suspension (UK SmPC) **[501]** Omeprazole 40 mg Powder for Solution for Infusion (UK SmPC) **[502]** Omeprazole, oral (US label) **[503]** Omeprazole, oral (US label, Bryant Ranch Prepack)

Full references p. 432–457